

AI/ML intern DAY 5

TASK 3 — ADVANCED STUDY OF VIRTUAL TRY-ON DATASETS

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1. Abstract

This task focused on researching datasets used in Artificial Intelligence virtual try-on systems and computer vision-based fashion applications. The study explored how AI training datasets improve pose estimation, clothing segmentation, garment alignment, and virtual fitting accuracy.

The research also analysed how datasets are structured, annotated, and processed for machine learning workflows in digital fashion technology.

2. Research Objective

The primary objective of this task was to study AI datasets used in virtual try-on systems and understand how machine learning models are trained using fashion-related data.

The research focused on:

- virtual try-on datasets
- pose estimation datasets
- clothing segmentation datasets
- fashion image annotation
- AI training workflows
- body landmark datasets

3. Introduction to AI Fashion Datasets

AI fashion datasets are structured collections of images and annotations used to train machine learning models for fashion-related computer vision applications.

These datasets help AI systems learn:

- body pose estimation
- clothing recognition
- human segmentation
- garment alignment
- outfit recommendation
- virtual fitting simulation

The datasets generally contain:

- human body images
- clothing images
- segmentation masks
- body landmarks
- pose annotations
- fashion labels



Figure 1 — AI Fashion Dataset Samples



4. Structure of Virtual Try-On Datasets

Most virtual try-on datasets contain multiple data components.

Dataset Component	Purpose
Human Images	Body pose analysis
Clothing Images	Garment simulation
Pose Annotations	Landmark detection
Segmentation Masks	Body region separation
Fashion Labels	Clothing categorization

These components help AI systems perform:

- pose tracking
- cloth alignment
- body segmentation
- clothing recommendation

5. Popular AI Fashion Datasets

Several datasets are widely used in virtual try-on and AI fashion applications.

5.1 DeepFashion Dataset

Figure 2 — DeepFashion Dataset

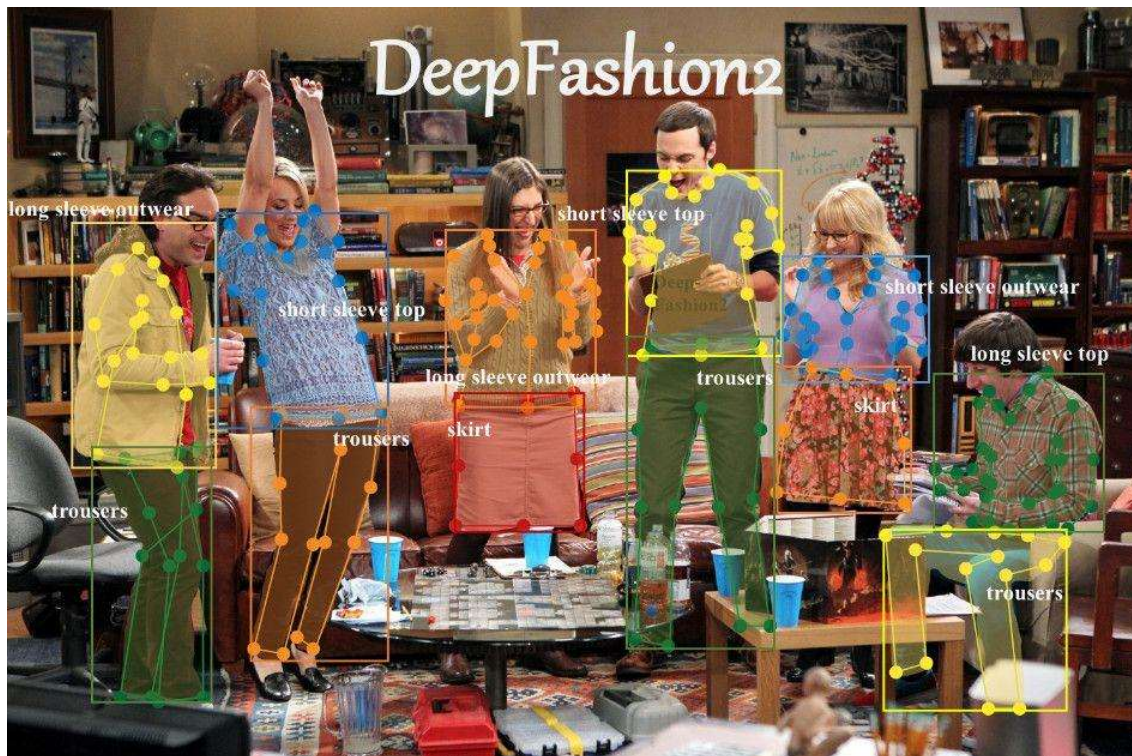




Figure 2. Example images of different categories and attributes in DeepFashion. The attributes form five groups: texture, fabric, shape, part, and style.

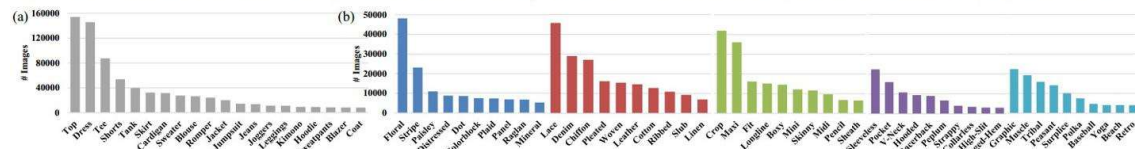


Figure 3. (a) Image number of the top-20 categories. (b) Image number of the top-10 attributes in each group.

Overview

DeepFashion is one of the largest fashion datasets used in computer vision and machine learning research.

Dataset Features

- over 800,000 fashion images
- clothing category labels
- pose annotations
- landmark annotations
- segmentation information

Applications

- clothing classification
- fashion recommendation
- virtual try-on systems
- clothing retrieval
- body landmark analysis

Advantages

- large dataset size

- detailed annotations
- high-quality fashion images

5.2 VITON Dataset

Figure 3 — VITON Virtual Try-On Dataset



Overview

VITON (Virtual Try-On Network) is a specialized dataset designed for AI clothing fitting and garment transfer systems.

Dataset Features

- paired clothing images
- body pose images
- garment alignment samples
- virtual fitting annotations

Applications

- virtual fitting rooms
- AI garment transfer
- digital clothing simulation
- pose-guided clothing generation

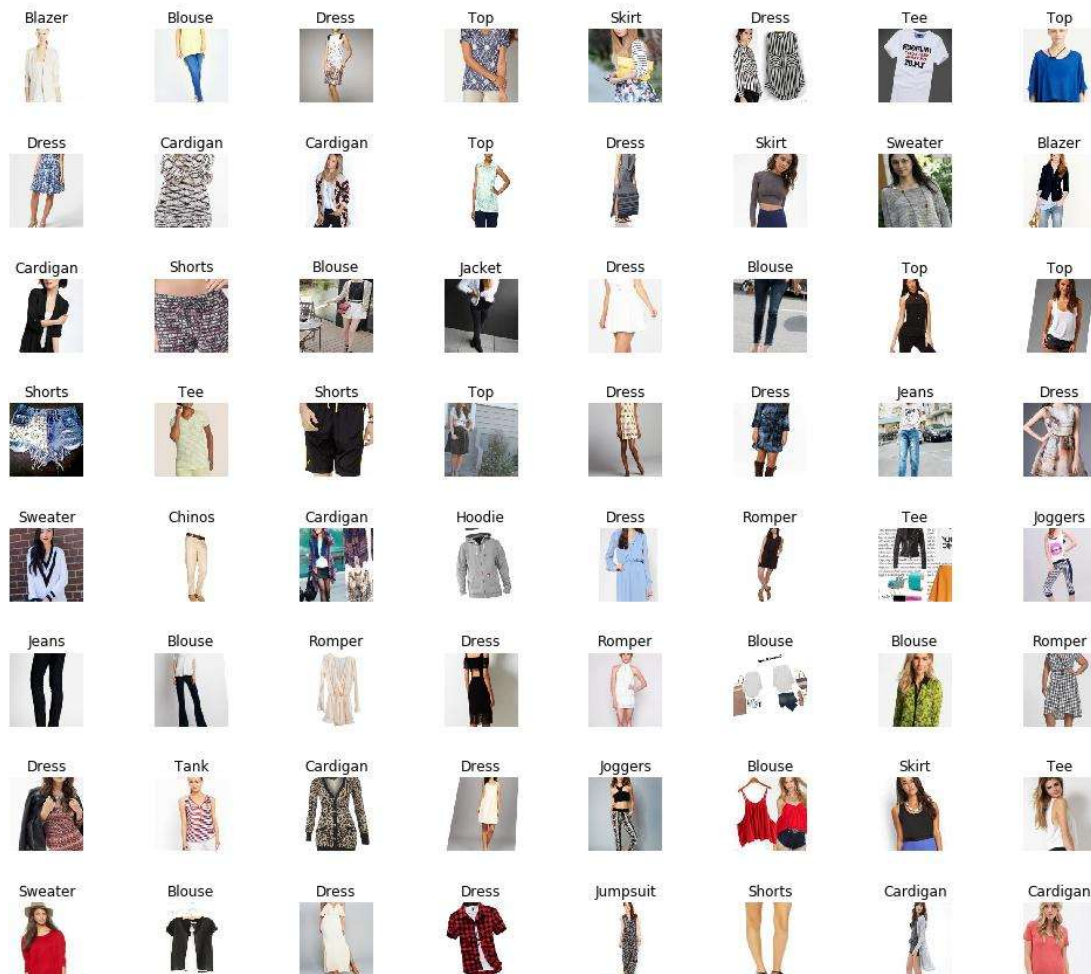
Advantages

- suitable for virtual try-on training
 - realistic garment fitting data
 - body-clothing alignment support
-

5.3 Fashion-MNIST Dataset

Figure 4 — Fashion-MNIST Dataset





Overview

Fashion-MNIST is a lightweight fashion dataset used for image classification and machine learning experiments.

Dataset Features

- grayscale clothing images
- labelled fashion categories
- small image dataset
- machine learning benchmark dataset

Clothing Categories

- shirts
- shoes
- bags

- trousers
- dresses

Applications

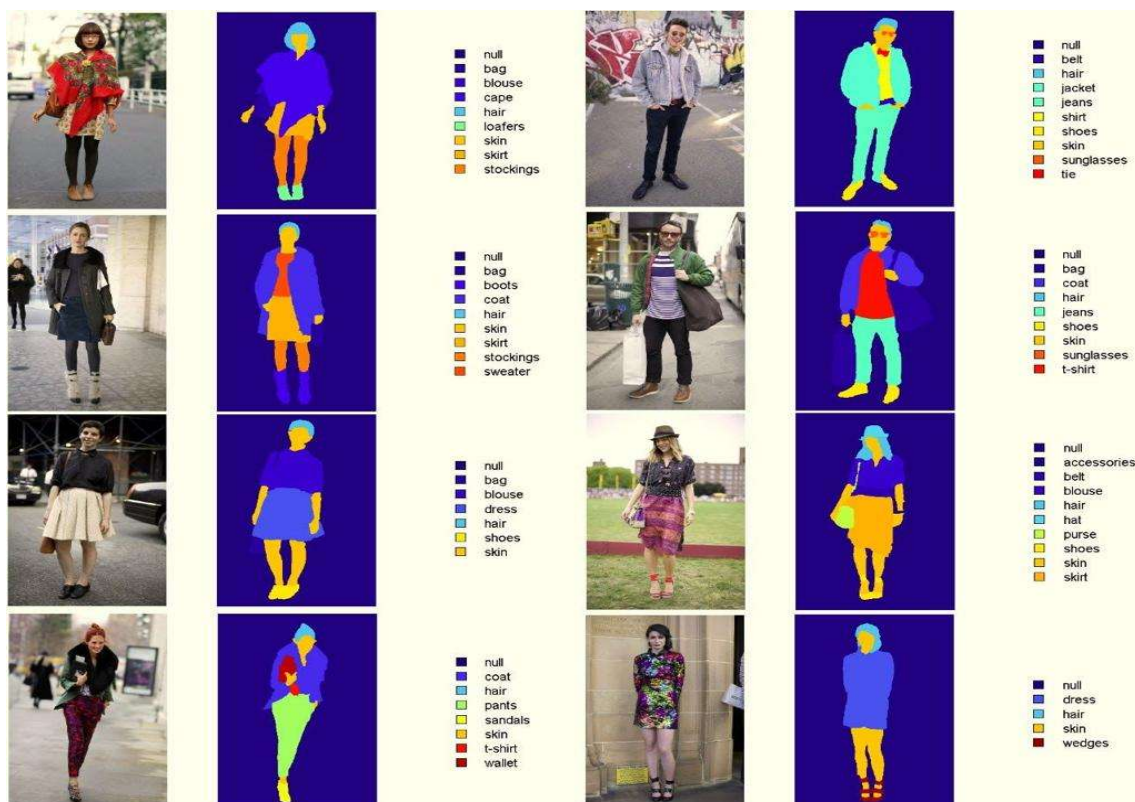
- image classification
- beginner machine learning projects
- fashion recognition systems

5.4 ATR Dataset

Figure 5 — ATR Human Parsing Dataset



Figure 7. Body-part segmentation with Sapiens-1B for 28 categories on single and multi-human images CSDN @咸鱼葵



Overview

The ATR Dataset is used for human parsing and body segmentation applications.

Dataset Features

- human body segmentation
- clothing region labels
- body parsing annotations
- foreground extraction masks

Applications

- body segmentation
- human parsing
- virtual fitting systems
- body region analysis

5.5 CIHP Dataset

Figure 6 — CIHP Human Parsing Dataset

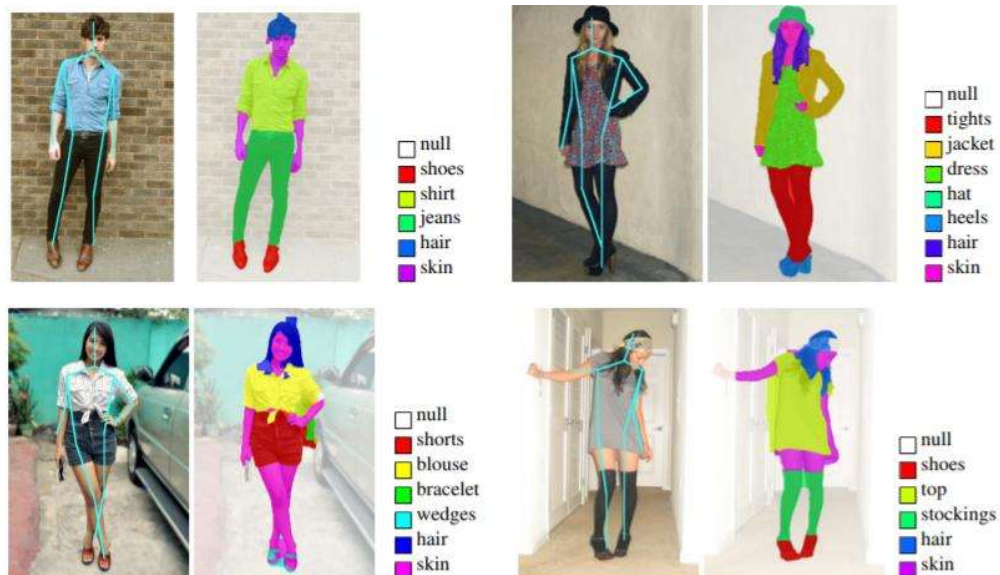
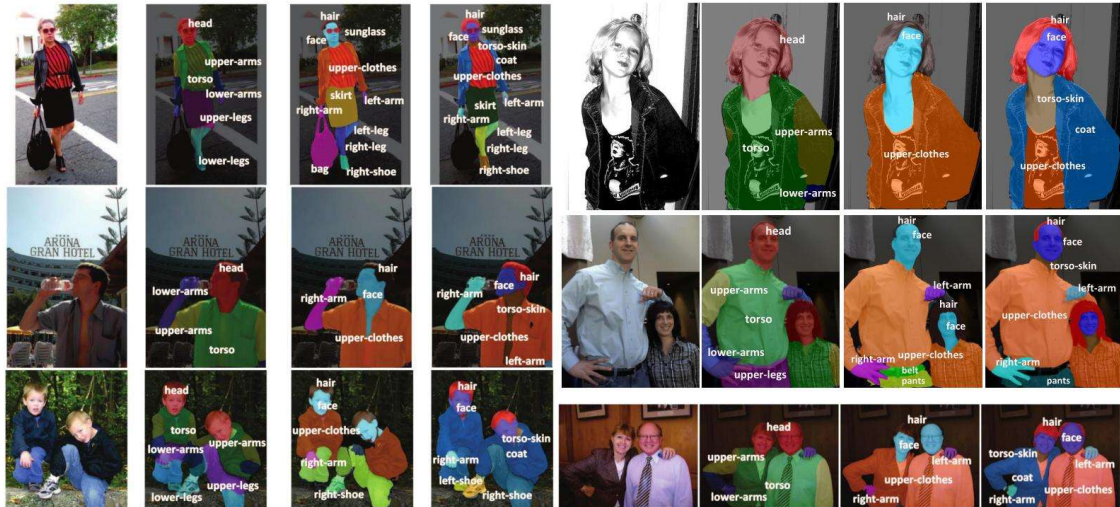


Figure 5: Example successful results on the Fashionista dataset.



Overview

CIHP (Crowd Instance-level Human Parsing) is a large-scale human segmentation dataset.

Dataset Features

- multiple human parsing
- segmentation masks
- body region annotations
- clothing parsing labels

Applications

- human segmentation
 - crowd body analysis
 - AI fashion applications
 - body tracking systems
-

6. AI Dataset Training Workflow

The AI training pipeline using fashion datasets generally follows several stages.

Step 1 — Data Collection

Fashion images and pose data are collected.

Step 2 — Data Annotation

Human body landmarks and clothing regions are labelled.

Step 3 — Data Preprocessing

Images are resized and normalized for AI training.

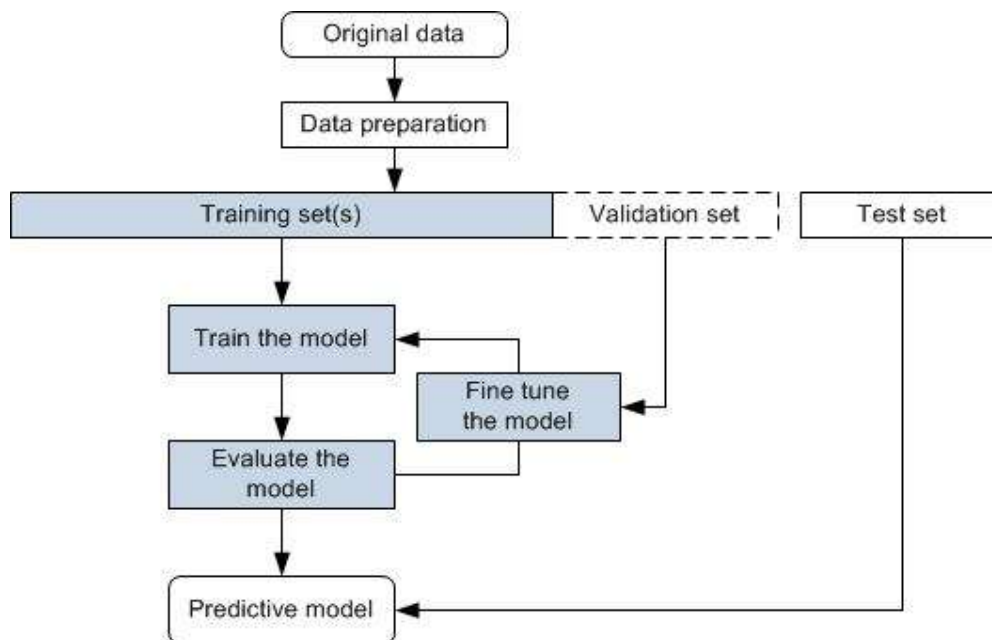
Step 4 — Model Training

Machine learning models learn clothing and body patterns.

Step 5 — Testing and Validation

The AI system is tested for prediction accuracy.

Figure 7 — AI Dataset Training Workflow



7. Advantages of AI Fashion Datasets

The advantages include:

- improved AI prediction accuracy
 - realistic garment simulation
 - better body segmentation
 - efficient AI training
 - improved clothing alignment
 - enhanced virtual fitting quality
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8. Challenges in Dataset Preparation

Challenge	Impact
Large dataset size	High storage requirement
Annotation complexity	Time-consuming labelling
Lighting variations	Affects training accuracy
Pose diversity	Requires multiple body positions
Dataset imbalance	Reduces model performance

9. Knowledge and Skills Acquired

During this research task, the following concepts were understood:

- AI dataset workflows
- fashion image annotation
- pose estimation datasets
- clothing segmentation systems
- machine learning training pipelines
- virtual try-on systems
- AI data preprocessing

10. Conclusion

Virtual try-on datasets are essential for training Artificial Intelligence systems used in modern fashion technology applications.

The research improved understanding of:

- AI fashion datasets
- dataset annotation systems
- clothing segmentation
- pose estimation workflows
- machine learning training
- digital fashion technologies

These datasets play a major role in improving AI virtual fitting systems, intelligent shopping platforms, and computer vision-based fashion applications.